

Ngoc Huynh

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EDUCATION

University of California, Irvine

Expected Graduation Date: Spring 2025

B.S. – Software Engineering, minor in Informatics and Computer Science

Relevant Coursework: Data Structures and Algorithms, Data Management, Computer Architecture, Operating Systems, Information Retrieval, Project in Databases and Web Applications, Software Testing and Analysis and Quality Assurance, Machine Learning and Data Mining, Internet Applications Engineering, Computer Networks, Embedded Systems, Software Design, Project Management

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, Kotlin, JavaScript, TypeScript, SQL, MATLAB, R, Lisp, Prolog, Swift

Software and Frameworks: HTML, CSS, Angular, React, Ionic, jQuery, Bootstrap, MySQL, PostgreSQL, Apache Cassandra, DataStax Astra, MongoDB, Couchbase, Neo4J, Apache Spark, Databricks, HTTP, API, Node.js, Express.js, Socket.io, NumPy, Jupyter Notebook, Streamlit, Visual Studio Code, IntelliJ IDEA, Git/GitHub, Android Studio, Xcode, AWS, Google Cloud, Figma, FigJam, WordPress

EXPERIENCE

Software Engineer Intern

January 2024 - June 2024

Zuum Transportation Inc

- Implemented a Robotic Process Automation System using Python, Gmail API for sending and receiving user emails, MongoDB to store user information, and integrated Zuum API to determine freight quotes based on key information extracted from emails.
- Leveraged natural language processing techniques such as Named Entity Recognition (NER), NLTK, and spaCy to accurately extract essential information from emails, and developed a Streamlit developer UI for future developers to test, annotate, and train the model for further improvement.

Student Accelerator

September 2022 - February 2024

California State University, Monterey Bay – Computing Talent Initiative

- Provided technical mentorship for 35-50 students, leveraging the UMPIRE method for coding interview problem-solving and facilitating weekly Deep Work Sessions on Data Structures, Algorithms, and Binary Numbers.
- Achieved top performance metrics, including the highest number of students in the green zone during Sprint 1 in Fall 2022 and Sprints 2 & 3 in Fall 2023, while organizing and accompanying students to tech events for skill enhancement and networking.

Undergraduate Researcher

August 2023 - December 2023

University of California, Irvine

- Conducted sentiment analysis on Yelp reviews using Neural Networks and Random Forest models to determine business experiences' positivity or negativity, with overall accuracy rate over 85%.
- Demonstrated the superiority of the Neural Networks model over Random Forest in accurately classifying sentiments. Emphasized the importance of refining tokenization strategies for enhanced sentiment classification in review-based platforms like Yelp.

Undergraduate Researcher, PI: Dr. Kiran George, Head of Computer Engineering

June 2022 - August 2022

California State University, Fullerton – Artificial Intelligence Lab

- Conducted an EEG-based study to investigate the impact of stress on Short-Term and Long-Term memory, utilizing advanced machine learning techniques such as SVM, KNN, and Neural Networks in MATLAB to analyze the data.
- Achieved high levels of accuracy ranging from 81.2% to 96.7% by developing and training sophisticated Neural Networks models and analyzed their performance using evaluation metrics such as confusion matrix and ROC curve. Utilized insights from the analysis to determine the most effective approach to detecting stress in the EEG data.

Project Lead Intern

March 2021 - August 2021

NASA/ California Space Grant Consortium

- Led a team of 2 other students to research about the Lunar Lava Tube Exploration and build an autonomous rover to find the possibility of human long-term habitation on the Moon.
- Utilized Arduino IDE and C language to control the movement of the rover and build a Graphic User Interface (GUI) by JavaScript to manipulate the rover's direction.

PROJECTS

- Travelog (April 2024 - June 2024):** Developed a travel website using React, Bootstrap, Node.js, Express.js, and MongoDB. Integrated Google Maps API for destination searches, Google Places API for detailed information, and OpenAI API for generating customizable travel plans. Enabled users to save and share plans on a social media feed.
- Fablix Movie Web App (April 2023 - May 2023):** Configured secure and responsive movie website hosting on AWS EC2, implementing Google ReCAPTCHA and HTTPS encryption. Utilized Tomcat server for frontend requests, retrieved JSON responses, and developed an Android application for seamless mobile access.
- RoommateFinder Web App | 1st place of MISSA ITC Web Application Development (April 2023):** Developed a responsive front-end using Angular and Bootstrap, while building a scalable back-end with Node.js, Express.js, and MongoDB. Implemented real-time chat features between users with Socket.io.